* Variables & Data Types

variable → Basically a container that hold hold value

How to declare variable

int firstnum = 10 ;

- int → Datatype (type of value)
- firstnum → varnam
- 10 → value

Datatypes in Java

DataType

- Primitive
  - ↳ Integer (byte, short, int, long)
  - ↳ Float (float, double)
  - ↳ character (char)
  - ↳ Boolean (boolean)
- non-primitive

Integer (byte, short, int, long)

- byte → 8 bit
- short → 16 bit
- int → 32 bit
- long → 64 bit

byte: −128 to +127 — firstnum = 10;

* Every no is signed (positive & negative repre) in Java

keywords - Reserved words (68 keywords)

* Types conversion in Java
  — converting a value from one data type to another

Datatypes
↳ Integers
↳ Real
↳ characters
↳ Booleans

Types conversion in Java
↳ Implicit (automatic done by compiler) (widening)
↳ Explicit (need to be done manually)

eg :

```
byte b = 24 ;
int i ;
i = b ;
s.o.pln (i)   //24
```

* Rules for Implicit conversion

- Destination datatype should be wider than source datatype.
- assigning smaller datatype to a variable of large datatype.
- short → int
  int → long

Explicit conversion (Narrowing)

- Destination datatype is small
- large data into smaller datatype

eg int i = 300 ;
byte b ;

b = i ; ✗ cannot be done

b = (byte) i ; ✓ done
cast

s.o.pln (b) ; → o/p 44

explanation : ovl of range -128 to 127

int i = 800 ;

Binary → 00101100

0000000	[illegible]	[illegible]	00101100
8	8	8	8

LSB

2	800
2	150 0
2	75
2	37
2	18
	9

32 16 8 4 2 1

[00101100]
8 bit

byte

1 1 1 1
16 + 8 + 4 + 2
= 44

code

```
public class Demo {
  public static void main (String[] args) {
    // Implicit conversion
    // (byte to int)
    byte b = 24 ;
      int i ;
      i = b ;
      s.o.pln (i)      // 24

    // Explicit conversion
      int i = 24 ;
      byte b ;
        b = (byte) i ;
        s.o.pln (b)      //24
```

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Automatic Type Promotion Rules

1. Any byte, short & char values are promoted to int
2. If one operand is long, the whole execution will become long
3. If one operand float will become float
4. If one operand is double, then entire execution will become double.

Program

```
byte b = 40;
char c = 'a';
short s = 1024;
int i = 50000;
float f = 5.67f;
double d = 0.1234;

double result = (f * b) + (i / c) - (d * s);

// f * b → float
// i / c → integer
// d * s → double

// float + integer → float - double → double

s.o.println (f * b) + " + " + (i / c) + " - " + (d * s));
s.o.println ("result =" + result);
```